

SMD IA 1 QUESTIONS

1. Question 1:

Explain the measures of central tendency and variability used in EDA. Give suitable examples.

Answer

Central Tendency

Central tendency gives a typical or central value around which the data is located.

1. Mean

The **mean** is the sum of all observations divided by the number of observations.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Example:

$$\{3, 5, 1, 2\}$$

$$\bar{x} = \frac{3 + 5 + 1 + 2}{4} = 2.75$$

The mean uses all observations but is **sensitive to extreme values (outliers)**.

 BAD702-module-1-textbook(5)

2. Median

The **median** is the middle value after sorting the data. It is also the **50th percentile**.

For even observations, it is the average of the two middle values.

The median is **robust**, because extreme values have little influence on it.

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3. Trimmed Mean

A **trimmed mean** is obtained by removing a fixed number of extreme values from both ends and calculating the mean of the remaining observations.

$$\bar{x}_{trim} = \frac{\sum_{i=p+1}^{n-p} x_i}{n - 2p}$$

It is a compromise between the mean and median and is more robust to extreme values.

Measures of Variability

Variability or dispersion measures how tightly clustered or spread out the observations are.

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1. Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

2. Variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

It measures the average squared deviation from the mean.

3. Standard Deviation

$$s = \sqrt{s^2}$$

It is easier to interpret because it is in the same units as the original data.

4. Mean Absolute Deviation

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

5. Median Absolute Deviation

It is the median of the absolute deviations from the **median**.

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Conclusion

Mean, median and trimmed mean describe **location**, while range, variance, standard deviation and absolute deviations describe **variability**. Mean and standard deviation are sensitive to outliers, whereas median and trimmed mean are more **robust**.

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2. Question 2:

Differentiate between Mean and Median.

Answer

Mean	Median
Sum of all values divided by number of values.	Middle value of sorted data.
Uses all observations .	Mainly depends on the central position .
Sensitive to outliers .	Robust to outliers .
Suitable when data is reasonably symmetric.	Useful for skewed data.
Formula: $\bar{x} = \frac{\sum x_i}{n}$	50th percentile.
Example For: 2, 3, 4, 5, 100 Mean: $\bar{x} = \frac{2 + 3 + 4 + 5 + 100}{5} = 22.8$ Median: 4 The extreme value 100 greatly increases the mean, while the median remains representative of the central data. Therefore, median is preferred when outliers or unusual distributions are present.	

3. Question

Explain the construction of a box plot from data. Identify potential outliers and explain why they appear in the box plot.

Answer

A **boxplot**, also called a **box-and-whiskers plot**, is a quick method for visualizing the distribution of numerical data. It is based on **percentiles**.

Construction

For the data:

2,4,5,6,7,8,9,10,12,20

First arrange the observations in ascending order.

Find:

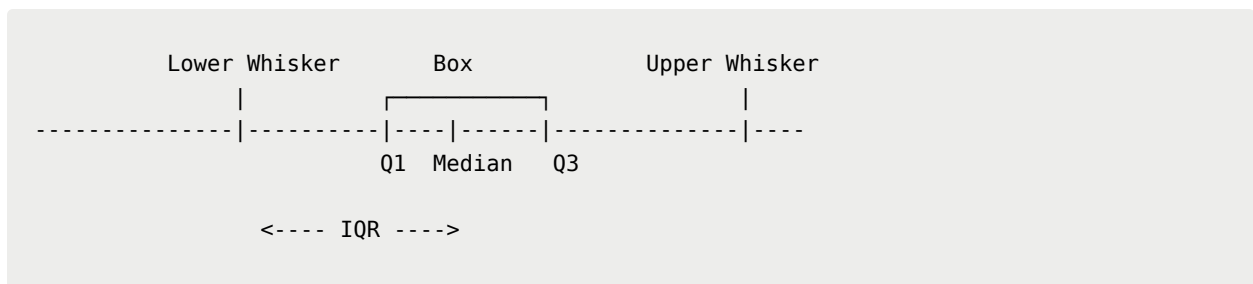
- $Q1 = 25\text{th percentile}$
- $Q2 = \text{median}$
- $Q3 = 75\text{th percentile}$

Then calculate:

$$IQR = Q3 - Q1$$

The box extends from $Q1$ to $Q3$, and the line inside the box represents the **median**.

Diagram



Identifying Outliers

Potential outliers are observations lying unusually far from the main body of the data.

A commonly used rule is:

$$\text{Lower limit} = Q1 - 1.5 (IQR)$$

$$\text{Upper limit} = Q3 + 1.5 (IQR)$$

Values outside these limits are shown separately from the whiskers as potential outliers.

Why do outliers appear?

An outlier is a value **very different from most of the data**. It may represent:

- A genuine extreme observation

- Data-entry or measurement error
- Sensor error
- Different units or populations

An outlier is **not automatically invalid** and should be investigated.

4. Question

Explain confidence intervals. How does a change in confidence level affect the width of the confidence interval? Also explain why confidence intervals are useful.

Answer

A **confidence interval (CI)** gives an **interval range** around a sample estimate to express uncertainty in the estimate.

The **confidence level** is the percentage of similarly constructed intervals expected to contain the statistic of interest.

General Form

$$\text{Estimate} \pm \text{Margin of Error}$$

For a mean using the normal approximation:

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

where:

- $\bar{x}$ = sample mean
- s = sample standard deviation
- n = sample size
- $z_{\alpha/2}$ = critical normal value.

Effect of Confidence Level

As the **confidence level increases**, the interval becomes **wider**.

For example:

$$90\% < 95\% < 99\%$$

Therefore:

$$\text{Higher confidence level} \Rightarrow \text{wider interval}$$

Similarly, a **smaller sample** produces a wider interval because there is greater uncertainty.

Importance

Confidence intervals:

1. Show the **uncertainty** associated with a point estimate.
2. Provide a range rather than a misleading single value.
3. Help determine whether a larger sample may be required.
4. Give information about the variability of a sample result.

5. Question

A sample of 100 observations has an average of 70 and standard deviation of 10. Construct a 95% confidence interval for the population mean.

Answer

Given

$$n = 100$$

$$\bar{x} = 70$$

$$s = 10$$

For a 95% confidence interval:

$$z_{0.025} = 1.96$$

Step 1: Calculate Standard Error

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$$

Step 2: Calculate Margin of Error

$$ME = 1.96 \times 1 = 1.96$$

Step 3: Construct Confidence Interval

$$CI = \bar{x} \pm ME$$

$$CI = 70 \pm 1.96$$

Therefore,

$$68.04 \leq \mu \leq 71.96$$

Final Answer

$$95\% \text{ CI} = (68.04, 71.96)$$

Thus, the population mean is estimated with a **95% confidence interval from 68.04 to 71.96.**

6. Question

Explain Binomial and Poisson distributions with suitable problems.

Answer

A. Binomial Distribution

The **binomial distribution** gives the distribution of the number of **successes** x in n trials, where every trial has two possible outcomes and probability p of success.

The probability of exactly x successes is:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Mean:

$$E(X) = np$$

Variance:

$$\text{Var}(X) = np(1 - p)$$

Example

If the probability of a sale from a click is $p = 0.1$, find the probability of exactly 2 sales in 5 clicks.

$$n = 5, \quad x = 2, \quad p = 0.1$$

$$\begin{aligned} P(X = 2) &= \binom{5}{2} (0.1)^2 (0.9)^3 \\ &= 10(0.01)(0.729) \end{aligned}$$

$$P(X = 2) = 0.0729$$

This is used for buy/don't buy, click/don't click, survive/die, etc.

 BAD702-module-2-textbook(1)

B. Poisson Distribution

The **Poisson distribution** models the number of events occurring randomly at a given overall rate λ during a specified time or space interval.

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

where:

- x = number of events
- λ = average number of events in the interval.

Applications

- Website visitors arriving over time
- Cars arriving at a toll plaza
- Typos per 100 lines of code
- Manufacturing imperfections.

 BAD702-module-2-textbook(1)

7. Question

Explain two-sample tests and the use of P-value or confidence interval in comparing two samples.

Answer

A **two-sample test** is used to determine whether there is a meaningful difference between two populations or treatment groups.

For example:

$$H_0 : \mu_A = \mu_B$$

$$H_1 : \mu_A \neq \mu_B$$

The difference between sample means is:

$$D = \bar{x}_A - \bar{x}_B$$

The **t-test** is commonly used for comparing two sample means. The t-distribution is used as a reference for differences between two sample means.  BAD702-module-2-textbook(1)

Procedure

1. State H_0 and H_1 .
2. Select significance level α .
3. Calculate the observed difference.
4. Calculate the test statistic.
5. Obtain the **P-value**.
6. Compare P-value with α .
7. Make the conclusion.

P-value decision

$$p \leq \alpha \Rightarrow \text{Reject } H_0$$

$$p > \alpha \Rightarrow \text{Fail to reject } H_0$$

Confidence Interval Method

A confidence interval for the difference can also be used.

If the interval for:

$$\mu_A - \mu_B$$

contains 0, there is not sufficient evidence of a difference at the corresponding confidence level.

If it does **not** contain 0, there is evidence of a difference.

Example from A/B testing

For two webpage groups, the notes give:

$$\bar{x}_A = 126.33$$

$$\bar{x}_B = 162.00$$

Thus:

$$D = 162.00 - 126.33 = 35.67$$

The question is whether this difference is within the range of what **random chance** could produce. A permutation test or two-sample t-test can be used.

 BAD702-module-3-textbook(1)

8. Question

Given a sample size of 11 patients, where the new treatment improves success by 0.1%, determine whether this difference is highly statistically significant and discuss practical significance.

Answer

Given

$$n = 11$$

Observed improvement:

$$0.1\%$$

The first point is that **statistical significance cannot be determined from the sample size and 0.1% improvement alone**. We would need sufficient information about the observed outcomes and variability, and then perform an appropriate statistical test.

A hypothesis test can be stated as:

H_0 : No difference between treatments

H_1 : There is a difference

The result would be judged using a **P-value** and a preselected significance level such as:

$$\alpha = 0.05$$

If:

$$p \leq 0.05$$

the result is statistically significant; otherwise, we fail to reject H_0 .

Practical Significance

An improvement of only:

$$0.1\%$$

is extremely small. Therefore, even if it were statistically significant, it may have **little practical significance**.

The notes emphasize that a statistically significant result does **not necessarily have practical significance**. A small difference can become statistically significant with sufficiently large samples, while significance alone does not measure the size or importance of an effect.

Conclusion

With only **11 patients** and a **0.1% improvement**, we **cannot claim high statistical significance without an actual hypothesis-test result**. Moreover, the effect is very small, so its **practical significance is likely to be negligible** unless the

application makes even a 0.1% improvement valuable. Statistical significance and practical importance must therefore be considered separately.
